TRAINITY PROJECT 2

Instagram User Analytics

Project Description:

- Analyze Instagram user interactions using SQL in MySQL Workbench to provide insights on engagement metrics (likes, comments, shares), user growth and retention, feature usage (Stories, Reels, DM), and content performance (videos, images, carousels). Also, explore demographic insights by segmenting users by location, age, and other factors. These findings will help the product and marketing teams make informed decisions to enhance user experience and guide the app's future development.

SQL Tasks:

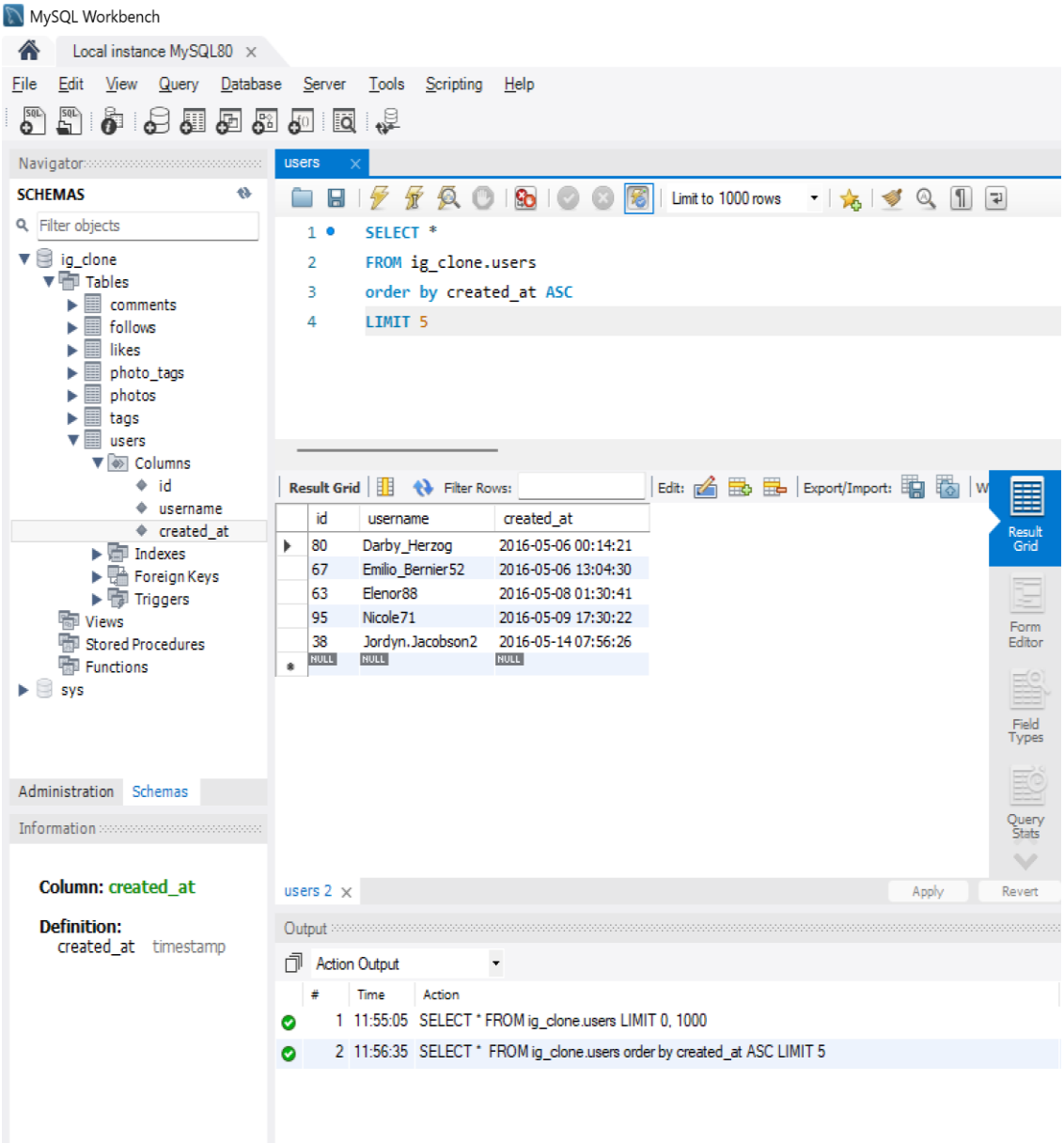
A) Marketing Analysis:

1. **Loyal User Reward:** The marketing team wants to reward the most loyal users, i.e., those who have been using the platform for the longest time.
   Your Task: Identify the five oldest users on Instagram from the provided database.
2. **Inactive User Engagement:** The team wants to encourage inactive users to start posting by sending them promotional emails.
   Your Task: Identify users who have never posted a single photo on Instagram.
3. **Contest Winner Declaration:** The team has organized a contest where the user with the most likes on a single photo win.
   Your Task: Determine the winner of the contest and provide their details to the team.
4. **Hashtag Research:** A partner brand wants to know the most popular hashtags to use in their posts to reach the most people.
   Your Task: Identify and suggest the top five most commonly used hashtags on the platform.
5. **Ad Campaign Launch:** The team wants to know the best day of the week to launch ads.
   Your Task: Determine the day of the week when most users register on Instagram. Provide insights on when to schedule an ad campaign.

Software Used: MySQL Workbench, Trainity Platform

Question 1: Identify the five oldest users on Instagram from the provided database.

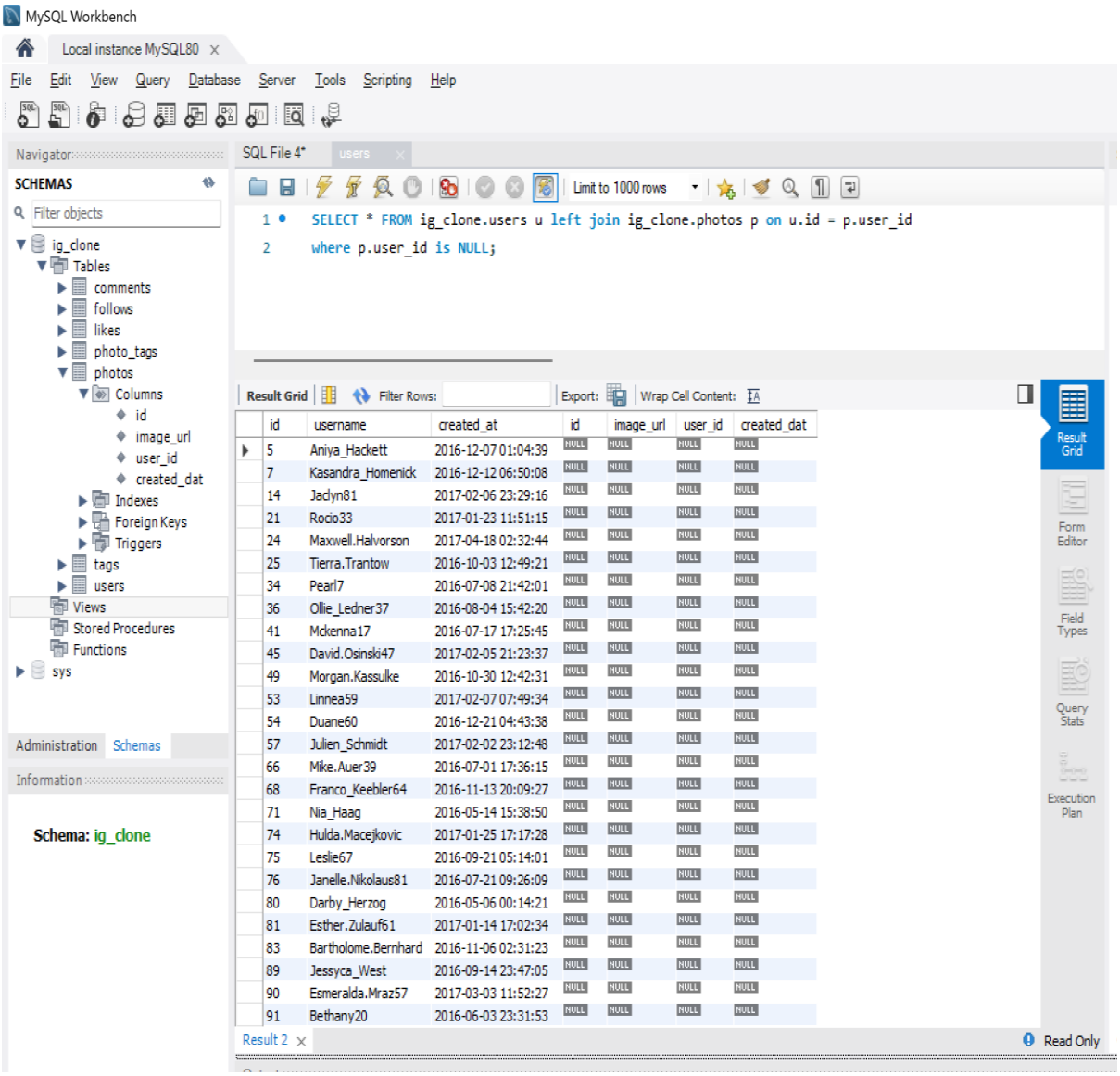
Ans:



- From the results, it can be seen that the five oldest Instagram customers are Darby Herzog, Emilio Berner 52, Elenor 88, Nicole 71 and Jordyn Jacobson 2.

Question 2: Identify users who have never posted a single photo on Instagram.

Ans:

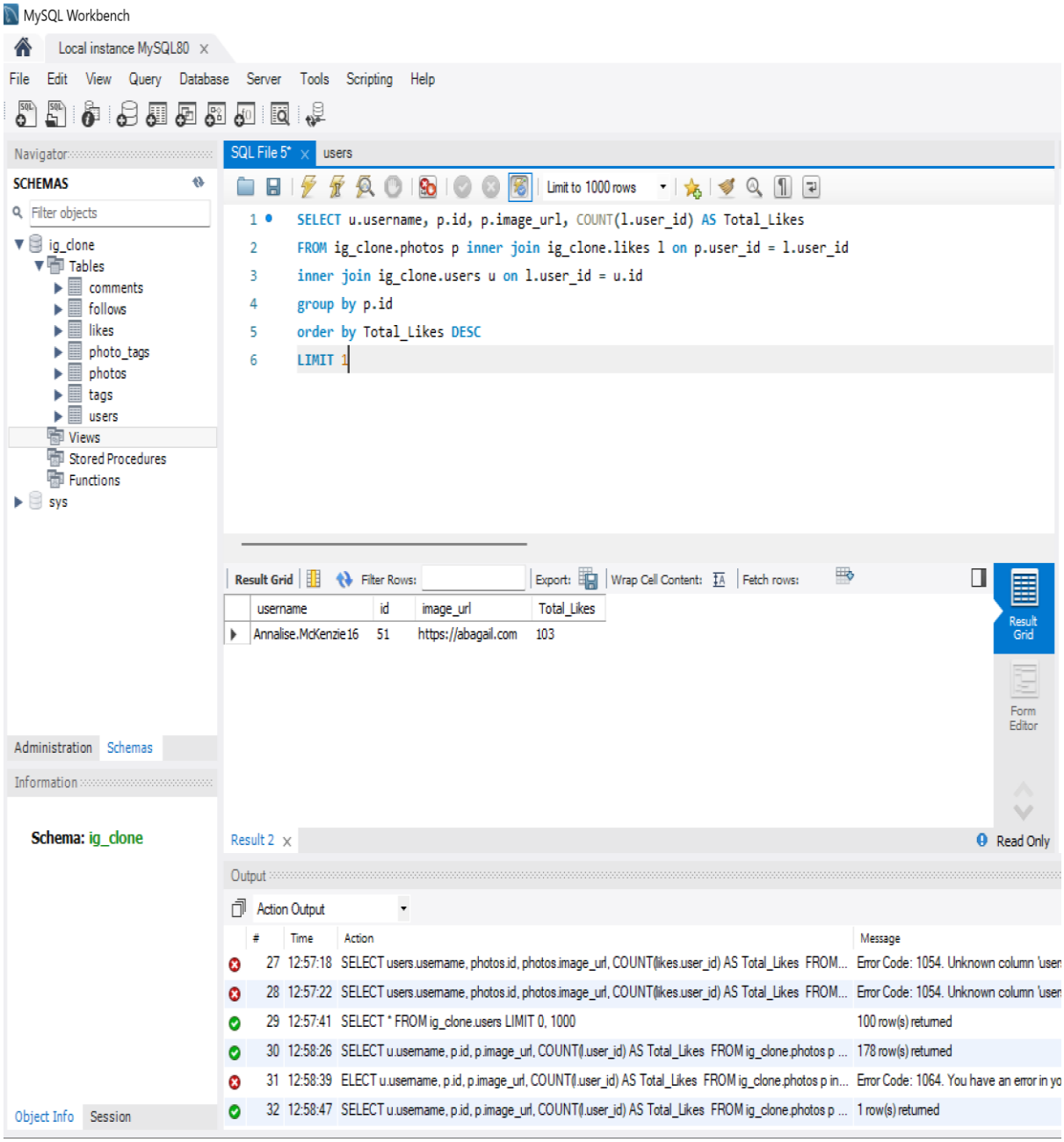


id	username	created_at	id	image_url	user_id	created_dat
5	Aniya_Hackett	2016-12-07 01:04:39	NULL	NULL	NULL	NULL
7	Kasandra_Homenick	2016-12-12 06:50:08	NULL	NULL	NULL	NULL
14	Jadyn81	2017-02-06 23:29:16	NULL	NULL	NULL	NULL
21	Rocio33	2017-01-23 11:51:15	NULL	NULL	NULL	NULL
24	Maxwell.Halvorson	2017-04-18 02:32:44	NULL	NULL	NULL	NULL
25	Tierra.Trantow	2016-10-03 12:49:21	NULL	NULL	NULL	NULL
34	Pearl7	2016-07-08 21:42:01	NULL	NULL	NULL	NULL
36	Ollie_Ledner37	2016-08-04 15:42:20	NULL	NULL	NULL	NULL
41	Mckenna17	2016-07-17 17:25:45	NULL	NULL	NULL	NULL
45	David.Osinski47	2017-02-05 21:23:37	NULL	NULL	NULL	NULL
49	Morgan.Kassulke	2016-10-30 12:42:31	NULL	NULL	NULL	NULL
53	Linnea59	2017-02-07 07:49:34	NULL	NULL	NULL	NULL
54	Duane60	2016-12-21 04:43:38	NULL	NULL	NULL	NULL
57	Julien_Schmidt	2017-02-02 23:12:48	NULL	NULL	NULL	NULL
66	Mike.Auer39	2016-07-01 17:36:15	NULL	NULL	NULL	NULL
68	Franco_Keebler64	2016-11-13 20:09:27	NULL	NULL	NULL	NULL
71	Nia_Haag	2016-05-14 15:38:50	NULL	NULL	NULL	NULL
74	Hulda.Macejkovic	2017-01-25 17:17:28	NULL	NULL	NULL	NULL
75	Leslie67	2016-09-21 05:14:01	NULL	NULL	NULL	NULL
76	Janelle.Nikolaus81	2016-07-21 09:26:09	NULL	NULL	NULL	NULL
80	Darby_Herzog	2016-05-06 00:14:21	NULL	NULL	NULL	NULL
81	Esther.Zulauf61	2017-01-14 17:02:34	NULL	NULL	NULL	NULL
83	Bartholome.Bernhard	2016-11-06 02:31:23	NULL	NULL	NULL	NULL
89	Jessyca_West	2016-09-14 23:47:05	NULL	NULL	NULL	NULL
90	Esmeralda.Mraz57	2017-03-03 11:52:27	NULL	NULL	NULL	NULL
91	Bethany20	2016-06-03 23:31:53	NULL	NULL	NULL	NULL

- The following users have never posted a single photo on Instagram. We have 26 users who have never posted a photo. Since we have 100 total users on Instagram, the inactive user rate is 26% which is good for a social media platform, though efforts to decrease this rate can we taken by launching more interactive feature and campaigns aligning with user engagement.

Question 3: Identify users who have never posted a single photo on Instagram.

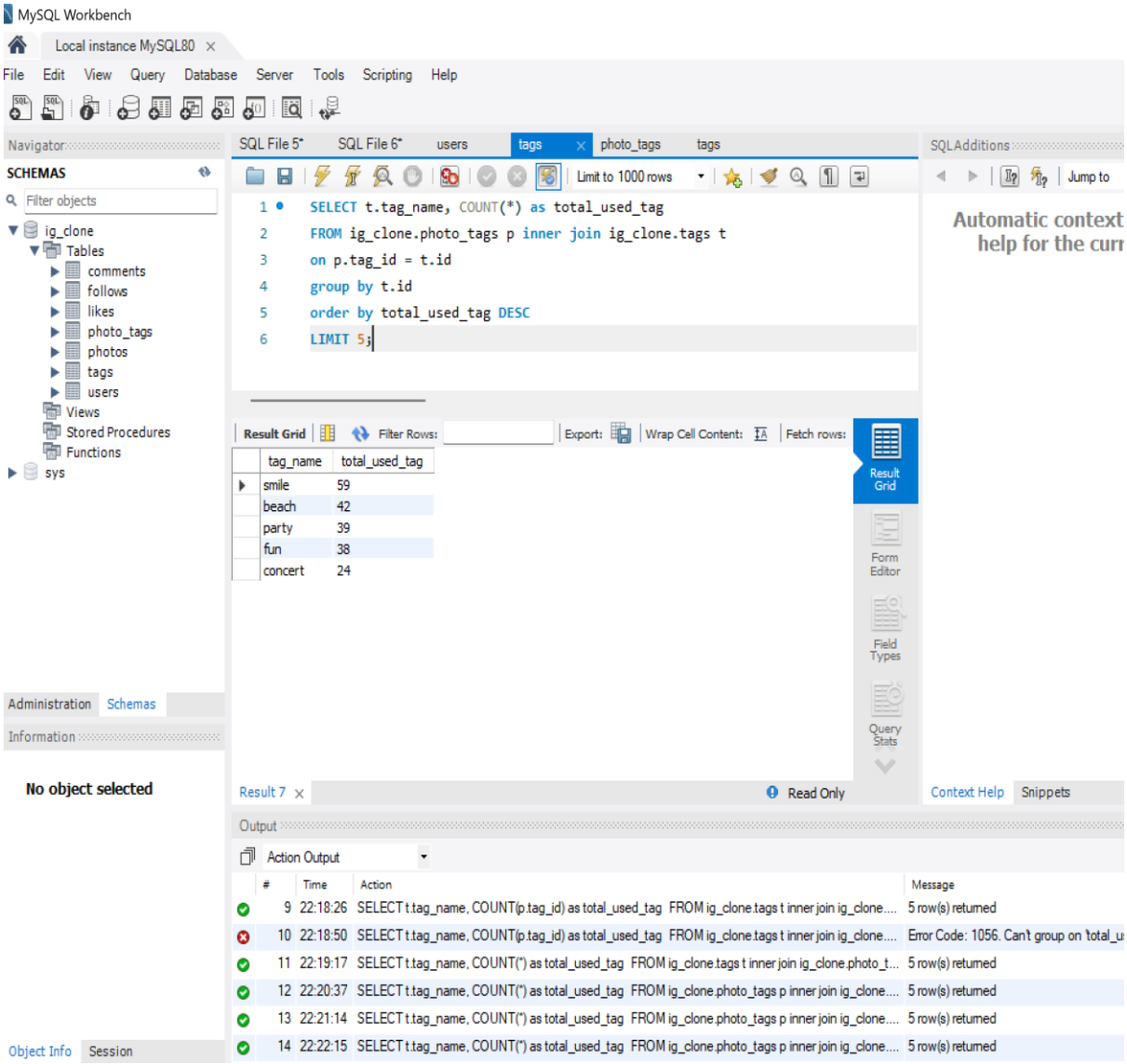
Ans:



- Annalise Mckenzie 16 has posted the photo with the greatest number of likes recorded to date. She could be hired as an ambassador for Instagram to further promote its engagement rate and get more users registered onto the platform.

Question 4: Identify and suggest the top five most commonly used hashtags on the platform.

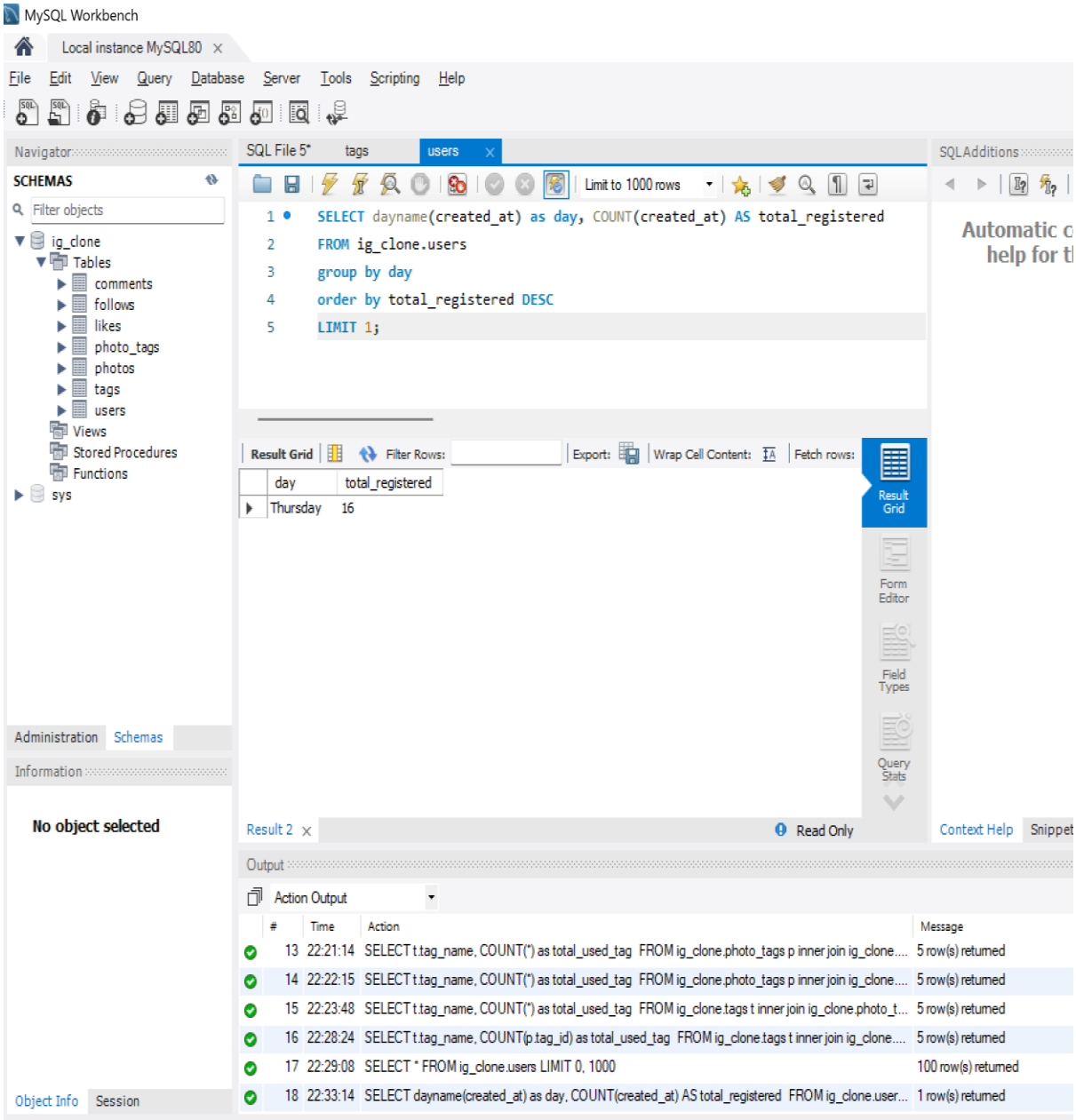
Ans:



- Smile, Beach, Party, Fun and Concert are the most used hashtags on the platform to date. Instagram could provide more features aligning with these hashtags as that could boost the engagement rate further.

Question 5: Determine the day of the week when most users register on Instagram. Provide insights on when to schedule an ad campaign.\

Ans:



- Based on the result, we have the most registration done a Thursday. As a result, the campaign should be launched on a Thursday, in-turn boosting this number further.

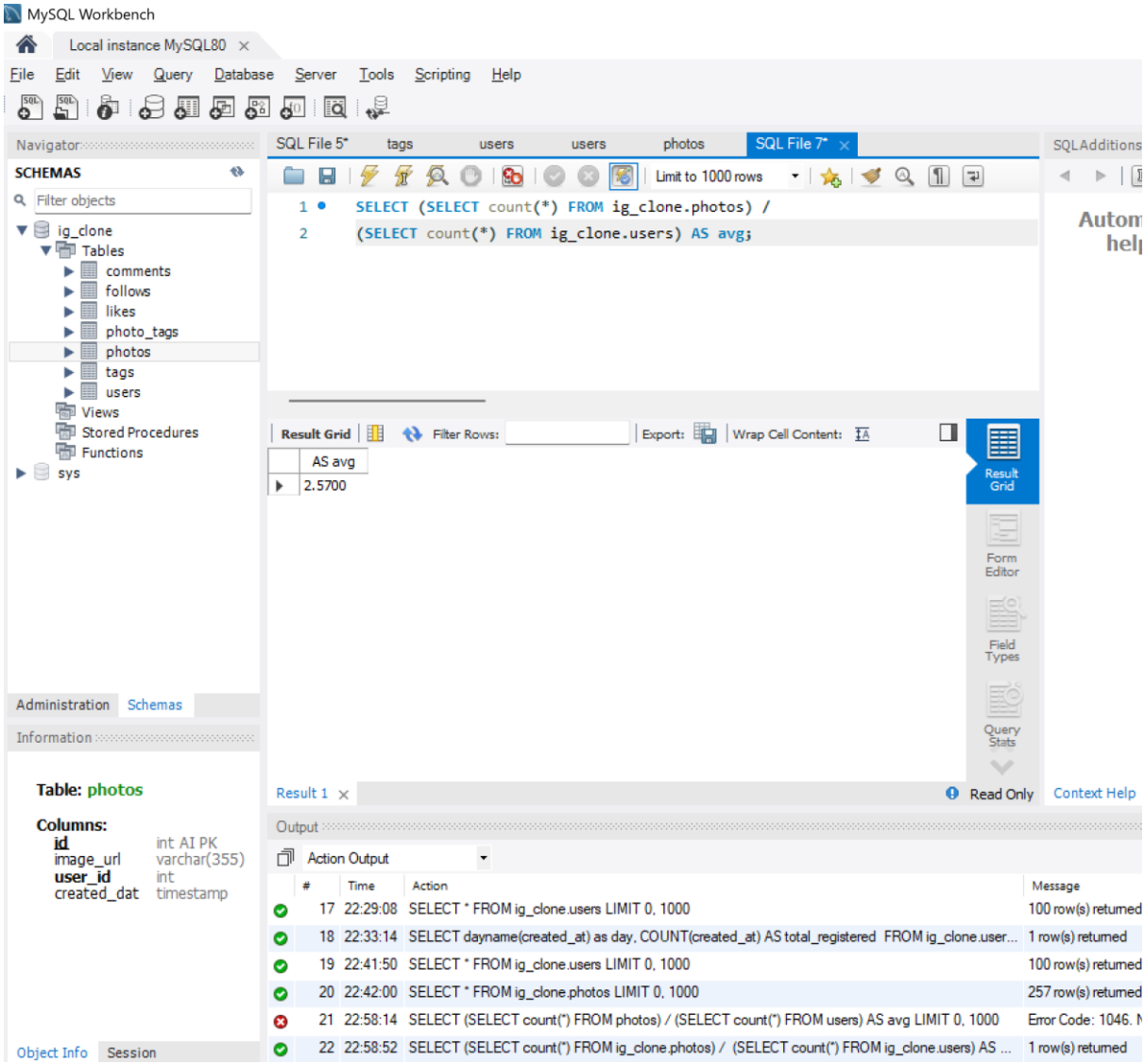
B) INVESTOR METRICS:

1. **User Engagement:** Investors want to know if users are still active and posting on Instagram or if they are making fewer posts.
   Your task: Calculate the average number of posts per user on Instagram. Also, provide the total number of photos on Instagram divided by the total number of users.

2. **Bots & Fake Accounts:** Investors want to know if the platform is crowded with fake and dummy accounts.
   Your task: Identify users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user.

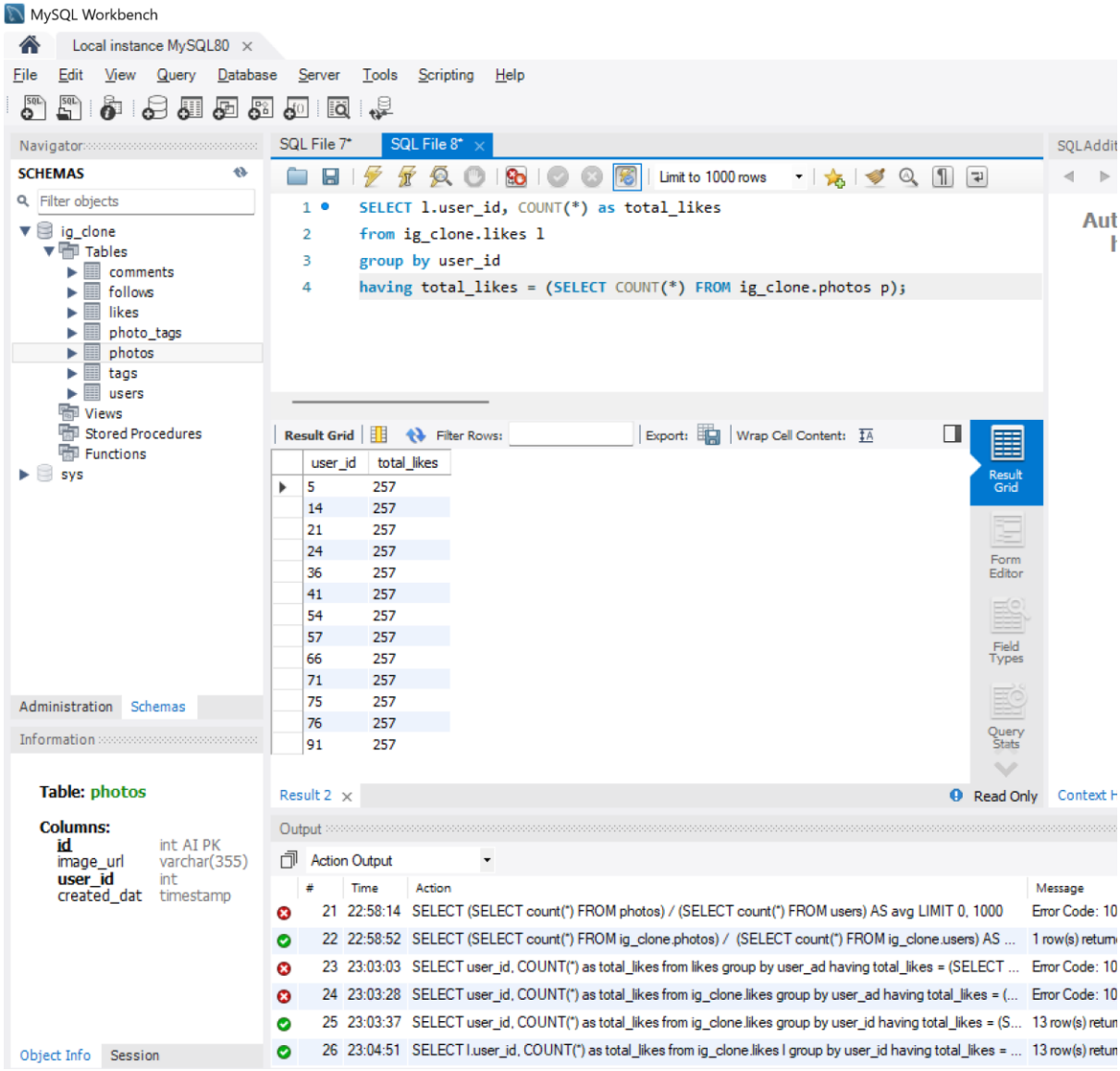
<u>Question 1:</u> Calculate the average number of posts per user on Instagram. Also, provide the total number of photos on Instagram divided by the total number of users.

Ans:



<u>Question 2:</u> Identify users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user.

Ans:



- We can see that we have a total of 13 bots of Instagram that have liked every single photo that has been posted on Instagram till date. This rate is higher for an application like Instagram. Therefore, steps to reduce this rate should be taken by the team.

Result: Through this project, I understood the basics of MySQL and how to write queries. I got a brief on the how to use SELECT, FROM, HAVING, GROUP BY, ORDER BY statements properly. I understood some concepts of using joins along with sub-queries as well while performing the “B” part of this project. I plan on further improving upon these concepts and writing more complex queries in the future.